Notes: Kwon and Tang (RES, 2026)

Extreme Categories and Overreaction to News

Contents

1 Big picture **3**

2 Data and measurement (key objects) **4**

2.1 Main datasets and sample construction . . . 4
2.2 Baseline categories and summary-statistic patterns . . . 4
2.3 Main constructed objects . . . 4
2.4 What the paper means by "extremeness" . . . 5

3 Models and empirical specifications (all equations + interpretation) **5**

3.1 Fundamentals with category-specific tails . . . 5
3.2 Diagnostic expectations and the reference distribution . . . 6
3.3 Asset demand and slow-moving arbitrage . . . 6
3.4 Main theoretical result for expectations . . . 7
3.5 Return predictability and trading volume . . . 7
3.6 How extremeness is estimated in the data . . . 8
3.7 Theoretical return relation and category-by-category drift/reversal regression . . . 8
3.8 Main pooled regression: reversals and extremeness . . . 9
3.9 Controlling for overlapping news . . . 9
3.10 Volume regression . . . 9
3.11 Alternative-explanations regression . . . 10

4 Identification and instruments (what makes the estimates credible) **10**

4.1 What the design is doing . . . 10
4.2 Main threats to credibility . . . 10
4.3 How the paper answers those threats . . . 11
4.4 What to keep in mind . . . 11

5 Tables and figures **11**

5.1 Figure 1: Diagnostic expectations and under/overreaction . . . 11

5.2 Figure 2: Theory – reversals and volume versus extremeness 12
5.3 Table 1: News category summary statistics 12
5.4 Figure 3: Estimating extremeness 12
5.5 Table 2: Category-level extremeness and drift/reversal 13
5.6 Table 3: Main regression of reversals on extremeness 13
5.7 Figure 4: Category-level scatter and horizon dynamics 14
5.8 Table 4: Robustness exercises 15
5.9 Table 5 and Figure 5: Volume and extremeness 15
5.10 Table 6 and Figure 6: Alternative explanations 16

6 Short summary **16**

7 Conclusion **16**

7.1 Core conclusion (what the paper establishes) 16
7.2 Economic intuition 17
7.3 Takeaway 17

1 Big picture

- **Question.** Why do investors seem to underreact to some news, such as earnings, but overreact to other news? The paper asks whether a systematic feature of the *news category itself* predicts whether prices drift or reverse after an announcement.
- **Core idea.** Investors interpret a news item by comparing it to similar past events. If a category contains very extreme outliers, that category is more likely to cue memorable tail events. Under diagnostic expectations, this makes investors overweight extreme states and overreact.
- **Theory.** The paper combines diagnostic expectations (representativeness-based belief formation) with category-specific power-law distributions for fundamentals. More extreme categories, meaning fatter-tailed categories, should generate more overreaction, more return reversals, and more disagreement-driven trading volume.
- **Key nuance.** The model predicts heterogeneity *across* categories, not within a category. Inside a given category, both small and large announcements are predicted to inherit the same average tendency toward underreaction or overreaction.
- **Data.** U.S. corporate news announcements from Capital IQ Key Developments, merged to CRSP daily returns and turnover, from 2011 to 2018. The baseline sample keeps firms listed on NYSE, AMEX, and NASDAQ with market capitalization above 2 billion dollars, and studies twenty-four fundamental corporate-news categories.
- **How extremeness is measured.** For each category, the paper estimates the tail exponent of the distribution of announcement-day absolute returns using a log-rank/log-value power-law regression. The inverse tail exponent, denoted ζ_C, is the paper's summary statistic for category extremeness.
- **Main empirical finding.** Extremeness strongly predicts post-announcement reversal. Least-extreme categories have modest drift, while most-extreme categories have sizable reversals. In the paper's quantitative summary, moving from the least- to the most-extreme categories shifts the post-announcement response from roughly a 6% drift to roughly a 24% reversal of the announcement-day move.
- **Additional finding.** Holding fixed announcement-day returns as a proxy for fundamentals, more extreme categories also have higher announcement-day turnover. The estimated increase is about 34% in turnover when moving from the least- to the most-extreme categories, conditional on a 10% announcement-day return.
- **Expectations evidence.** In supplementary evidence using analyst EPS forecasts, forecast errors are more negatively predicted by revisions for more extreme categories, which is consistent with the diagnostic-expectations mechanism.
- **Why this paper matters.** It offers a concrete, measurable determinant of overreaction. Instead of asking only whether *big* news reverses, the paper asks whether news from categories with more extreme tails reverses more. That connects a behavioral theory of salience and representativeness to a statistic that can be estimated directly from the data.

2 Data and measurement (key objects)

2.1 Main datasets and sample construction

- **News data.** Capital IQ Key Developments, which codes firm-level corporate announcements such as earnings, earnings calls, product launches, client news, leadership changes, lawsuits, business expansions, dividend news, and mergers and acquisitions.
- **Returns and volume.** CRSP daily stock returns and trading volume.
- **Media intensity.** RavenPack, used later when the paper asks whether media coverage, rather than extremeness, explains the results.
- **Forecast data.** I/B/E/S analyst EPS forecasts are used in supplementary tests of forecast error predictability.
- **Sample period.** 2011–2018.
- **Firms.** U.S. firms on major exchanges (NASDAQ, NYSE, AMEX).
- **Baseline market-cap filter.** Firms below 2 billion dollars in market capitalization are excluded to reduce concerns that very short-run reversals are driven by microstructure and liquidity effects rather than investor psychology.
- **Category filter.** The baseline sample keeps categories with at least 1,000 observations and removes categories judged not to be directly about firm fundamentals, such as administrative filings, trading-status events, and issuance/repurchase events like IPOs and SEOs.
- **Timing rule.** After-hours announcements are matched to the next trading day.

2.2 Baseline categories and summary-statistic patterns

- **High-frequency categories.** Earnings (23,573 announcements), product news (23,500), board changes (22,173), client news (20,990), and guidance confirmations (19,528) are among the largest categories.
- **No-news benchmark.** The paper keeps 2,801,102 no-event firm-days to characterize the reference distribution of returns when no public announcement occurs.
- **Return volatility on announcement days.** Most categories have standard deviations above 2.1%, which is roughly the sample's average daily volatility on no-news days. Earnings-call, guidance, and credit-watch announcements are especially volatile.
- **Turnover.** Announcement days also have meaningfully elevated turnover. For example, average turnover is around 2.20 for earnings calls, 2.53 for guidance-lower announcements, and 2.19 for credit-watch news, versus 0.99 on no-event days.
- **Interpretation.** News days are both high-volatility and high-volume days, which makes them a natural setting to study whether investor beliefs overshoot or undershoot.

2.3 Main constructed objects

- **Announcement-day return.** $r_{i,t,C}$ is the firm's stock return on the trading day associated with news item $n_{i,t,C}$.

- **Post-announcement return.** $r_{i,t+1,t+k,C}$ is the cumulative return over the next k trading days. The baseline horizon is $k = 90$.
- **Extremeness.** ζ_C is the inverse power-law index of the tail distribution of announcement-day absolute returns for category C.
- **Volume.** Turnover on the announcement day, measured as shares traded times share price divided by market capitalization, times 100.
- **Alternative extremeness measures.** The paper also computes extremeness using longer-run stock returns and realized earnings growth over horizons of one to five years.
- **Alternative category-level characteristics.** Informativeness, media coverage, the mean and dispersion of announcement-day returns, the number of occurrences, the mean of the five largest announcement returns, and whether the category is scheduled in advance.

2.4 What the paper means by "extremeness"

- **Not variance.** Extremeness is not just a high standard deviation.
- **Not average size.** It is also not the mean absolute return in the category.
- **What it captures.** It measures how far the tail sits from the typical event in that category. A category is extreme when its top 1% events are much larger than its ordinary events.
- **Illustrative example.** Earnings announcements often generate large announcement-day returns, but they are among the *least* extreme categories in the tail sense because the best earnings events are not wildly more consequential than other positive earnings events.

3 Models and empirical specifications (all equations + interpretation)

3.1 Fundamentals with category-specific tails

The stock has initial fundamentals $F_0 > 0$, and future fundamentals are

$$F_{\text{final}} = \exp(v) F_0. \tag{1}$$

If no public announcement occurs, fundamentals follow a symmetric density

$$\pi(v \mid d) \equiv \begin{cases} \pi_{0,d}(v) & \text{for } |v| < v_{0,d}, \\ C \cdot |v|^{-(\zeta_d^{-1}+1)} & \text{for } |v| > v_{0,d}. \end{cases} \tag{2}$$

If instead news from category $C \in \mathcal{C}$ arrives, fundamentals are power-law distributed:

$$\pi(v \mid C) = \frac{\zeta_C^{-1}}{2} v_{0,C}^{\zeta_C^{-1}} |v|^{-(\zeta_C^{-1}+1)} \qquad \text{for } |v| \geq v_{0,C}. \tag{3}$$

Conditional on receiving an idiosyncratic signal $s_j = v \cdot u_j$ with $u_j \sim \text{Unif}[0,1]$ for $v > 0$, the rational posterior is

$$\pi(v \mid C, s_j) = (\zeta_C^{-1} + 1) v_{1,C}^{\zeta_C^{-1}+1} v^{-(\zeta_C^{-1}+2)} \qquad \text{for } v \geq v_{1,C} = \max\{s_j, v_{0,C}\}. \tag{4}$$

- **Interpretation.** Each news category has its own tail thickness. The bigger is ζ_C, the fatter is the tail and the more important are rare outlier events in that category.
- **Economic meaning.** Categories such as M&A rumors or CEO changes may contain rare events with huge implications for fundamentals, while categories such as earnings or guidance confirmations may be more regular and less tail-driven.
- **Key modeling choice.** Extremeness is a property of the *category distribution*, not of a specific announcement's realized size.

3.2 Diagnostic expectations and the reference distribution

The general diagnostic-expectations distortion is

$$\pi_\theta(v \mid C, s_j) \propto \pi(v \mid C, s_j) \cdot \left(\frac{\pi(v \mid C, s_j)}{\pi_0(v)}\right)^\theta, \tag{5}$$

where $\theta \geq 0$ governs the strength of representativeness. The "no-news" reference distribution is

$$\pi_0(v) = \pi(v \mid s = 0) \propto \left(p_d \pi(v \mid d) + \sum_{C \in \mathcal{C}} p_C \pi(v \mid C)\right) \cdot \frac{1}{v}. \tag{6}$$

The paper's main specification instead conditions the reference on no public announcement:

$$\pi_\theta(v \mid C, s_j) \propto \pi(v \mid C, s_j) \cdot \left(\frac{\pi(v \mid C, s_j)}{\pi(v \mid d, s_j = 0)}\right)^\theta. \tag{7}$$

The parameter space is restricted so that diagnostic expectations remain well-defined:

$$\eta_{\min} \equiv \frac{\sqrt{1+\theta}}{\sqrt{1+\theta} + \sqrt{1+\zeta_d}} < 1, \qquad \eta_{\max} \equiv \frac{1+\theta}{\theta} > 1, \qquad \eta_{\min}\zeta_d \leq \zeta_C \leq \eta_{\max}\zeta_d. \tag{8}$$

- **Interpretation.** Investors overweight states that have become disproportionately more likely in light of the news. The reference distribution is "what would have happened without this kind of news."
- **Why tails matter.** If the category has a fatter right tail than the reference, then extreme positive outcomes become especially representative after a good signal, and diagnostic investors overshoot.
- **Thin-tail case.** If the category is less extreme than the reference, the news is more representative of middling outcomes, and investors underreact.

3.3 Asset demand and slow-moving arbitrage

Diagnostic and rational investors demand the asset according to subjective expected log returns:

$$D_j^{DE}(s_j, p) = \kappa\Big(\mathbb{E}_{\theta,j}[\log(F_{\text{final}})] - \log p\Big), \qquad D_j^{RE}(s_j, p) = \kappa\Big(\mathbb{E}_j[\log(F_{\text{final}})] - \log p\Big). \tag{9}$$

- **Timing.** At $t = 1$ only diagnostic investors trade, so prices initially reflect distorted beliefs. At $t = 2$ a large mass of Bayesian investors arrives, and prices converge to rational valuations.
- **Interpretation.** This is a slow-moving-arbitrage environment. Behavioral traders move prices first; rational traders correct them later.
- **Role of heterogeneous signals.** Different investors draw different s_j, which generates disagreement and thus trading volume on the announcement day.

3.4 Main theoretical result for expectations

Proposition 1 shows that diagnostic expectations are a scalar multiple of rational expectations:

$$\mathbb{E}_\theta[v \mid C, s_j] = \psi(\zeta_C, \zeta_d, \theta) \cdot \mathbb{E}[v \mid C, s_j] = \frac{1 + \zeta_C + \theta\left(1 - \frac{\zeta_C}{\zeta_d}\right)}{1 + \zeta_C + (1 + \zeta_C)\theta\left(1 - \frac{\zeta_C}{\zeta_d}\right)} \cdot \mathbb{E}[v \mid C, s_j]. \tag{10}$$

The paper then maps this into a forecast-error regression of the Coibion–Gorodnichenko type:

$$FE_{j,t=1} = \alpha + \beta_C^{CG} FR_{j,t=1} + \varepsilon_j, \tag{11}$$

where $FE_{j,t=1} = v - \mathbb{E}^\theta_{j,t=1}[v \mid C, s_j]$ and $FR_{j,t=1} = \mathbb{E}^\theta_{j,t=1}[v \mid C, s_j]$.

- **Overshoot condition.** $\psi > 1$ if and only if $\zeta_C > \zeta_d$. So diagnostic expectations overshoot the rational benchmark exactly when the category is more extreme than the no-news reference distribution.
- **Forecast implication.** β_C^{CG} decreases in ζ_C and is negative precisely when the category is more extreme than the reference. More extreme categories therefore generate more overreaction in expectations.
- News in an extreme category feels like "this could be the next huge winner," so investors put too much probability on tail outcomes.

3.5 Return predictability and trading volume

Define the diagnostic tail parameter by

$$\zeta_{C,\theta}^{-1} \equiv \zeta_C^{-1} + \theta(\zeta_C^{-1} - \zeta_d^{-1}), \qquad \eta_C(v) \equiv \frac{v_{0,C}^2 + v^2}{2v}. \tag{12}$$

Then period-1 and period-2 returns satisfy

$$r_2 = \beta_C^{\text{ret}} \cdot r_1, \qquad \beta_C^{\text{ret}} \equiv \frac{\zeta_C - \zeta_{C,\theta}}{1 + \zeta_{C,\theta}}. \tag{13}$$

Announcement-day trading volume is

$$Vol = \frac{1}{2}\kappa\,(1 + \zeta_{C,\theta})\,(1 - \eta_C(v))^2. \tag{14}$$

- **Interpretation of β_C^{ret}.** This is the theoretical drift/reversal coefficient linking announcement-day returns to subsequent returns. If $\beta_C^{\text{ret}} < 0$, prices reverse and the announcement-day response was an overreaction. If $\beta_C^{\text{ret}} > 0$, prices drift and the initial response was an underreaction.
- **Comparative statics.** β_C^{ret} decreases in both ζ_C and θ. More extreme categories and stronger diagnostic distortions imply more reversal.
- **Volume implication.** Holding fundamentals fixed, more extreme categories produce more disagreement across investors and therefore more turnover.
- **Cross-category benchmark.** The sign switch occurs at $\zeta_C = \zeta_d$. The no-news tail is therefore the relevant benchmark that separates underreaction from overreaction.

3.6 How extremeness is estimated in the data

The paper's main extremeness estimator uses the top decile of absolute announcement-day returns:

$$\log(\text{Rank}_{i,t,C} - 0.5) = \xi_C - \hat{\zeta}_C^{-1} \log(|r_{i,t,C}|), \qquad |r_{i,t,C}| > |r_{C,90}|. \tag{15}$$

The paper also measures fundamentals using realized EPS growth:

$$\hat{v}(e_{i,t,C})_{EPS,k} = \frac{EPS_{i,t-1+k,C}}{EPS_{i,t-1,C}} - 1, \qquad 1 \leq k \leq 5. \tag{16}$$

- **Interpretation.** In the rank-size regression, a higher $\hat{\zeta}_C$ means that large tail observations decline more slowly with rank, so the category is more fat-tailed.
- **Why announcement-day returns are the baseline.** They are closely tied to the specific news item and are available for every announcement, while long-run fundamentals are noisier and harder to attribute cleanly to one event.
- **Prediction 4.** If the theory is right, extremeness estimated from announcement-day returns, longer-run returns, and realized earnings growth should all be positively correlated across categories.

3.7 Theoretical return relation and category-by-category drift/reversal regression

The theoretical relation is

$$r_2 = \beta_C^{\text{ret}}(\zeta_C, \theta, \zeta_d) \cdot r_1. \tag{17}$$

The category-specific empirical counterpart is

$$r_{i,t+1,t+k} = \alpha + \beta_C r_{i,t} + \varepsilon_{i,t}. \tag{18}$$

To test whether drift and reversal differ across categories, the paper estimates

$$r_{i,t+1,t+k,C} = \alpha + \sum_{C \in \mathcal{C}} \beta_C \mathbf{1}(News_C) \cdot r_{i,t,C} + \mu_C + \varepsilon_{i,t,C}. \tag{19}$$

- **Interpretation.** β_C is the empirical drift/reversal coefficient for category C.
- **Reading the sign.** A positive β_C means drift; a negative β_C means reversal.
- **Main empirical fact.** The paper rejects both that all β_C are equal to zero and that all β_C are equal to each other.

3.8 Main pooled regression: reversals and extremeness

The core pooled specification is

$$r_{i,t+1,t+k,C} = \alpha + \beta r_{i,t,C} + \gamma\,\zeta_{C,t} \times r_{i,t,C} + \varepsilon_{i,t,C}. \tag{20}$$

- **Coefficient of interest.** γ measures whether more extreme categories have more drift or more reversal.
- **Prediction.** $\gamma < 0$. Extremeness should tilt the response toward reversal.
- **Interpretation of magnitude.** The post-announcement slope for a given category is $\beta + \gamma\zeta_C$. Hence higher ζ_C lowers the continuation of announcement-day returns.

3.9 Controlling for overlapping news

To address the concern that one category's reversals may mechanically reflect reactions to earlier announcements, the paper estimates

$$\begin{aligned} r_{i,t+1,t+k,C} = \alpha &+ \beta_0 r_{i,t,C} + \gamma\,\zeta_{C,t} \times r_{i,t,C} \\ &+ \sum_{C' \in \mathcal{C}} \sum_{h=1}^{90} \theta_{C',h}\, I(i, C', t-h)\, r_{i,t-h,C'} + \varepsilon_{i,t,C}. \end{aligned} \tag{21}$$

- **Interpretation.** The regression partials out the effect of all prior announcements over the preceding ninety days.
- **Why this matters.** If, say, earnings drift continues for a while, a later news category could inherit that drift mechanically. This specification removes that channel.

3.10 Volume regression

To test Prediction 3, the paper estimates

$$Turnover_{i,t,C} = \alpha + \beta|r_{i,t,C}| + \delta|r_{i,t,C}| \cdot \zeta_C + \mu_t + \mu_C + \varepsilon_{i,t,C}. \tag{22}$$

It also computes category-specific turnover conditional on a 10% absolute announcement return from

$$Turnover_{i,t,C} = \alpha_C + \sum_{C \in \mathcal{C}} T_C \mathbf{1}(News_C) \cdot |r_{i,t,C}| + \varepsilon_{i,t,C}. \tag{23}$$

- **Coefficient of interest.** $\delta > 0$ means that more extreme categories generate more trading even after conditioning on the magnitude of the return response.

- **Economic reading.** Investors disagree more when a category is strongly associated with salient tail outcomes.

3.11 Alternative-explanations regression

To compare extremeness with competing stories, the paper estimates

$$r_{i,t+1,t+k,C} = \alpha + \beta r_{i,t,C} + \gamma\, \zeta_C\, r_{i,t,C} + \phi X_C r_{i,t,C} + \varepsilon_{i,t}, \tag{24}$$

where X_C is one alternative category-level characteristic such as informativeness, media intensity, scheduled timing, the mean return, the interquartile range, the standard deviation, occurrence counts, or the size of the largest announcements.

- **Interpretation.** This asks whether extremeness still predicts reversal once other plausible category-level explanations are allowed to vary with announcement-day returns.
- **Empirical result.** Across all these horse races, extremeness remains statistically and economically important.

4 Identification and instruments (what makes the estimates credible)

4.1 What the design is doing

- **No formal instrument.** This is not an IV paper. Identification comes from a theory-guided cross-category comparison: categories with different measured tail thickness should exhibit systematically different post-announcement dynamics.
- **Key empirical object.** The main object is the interaction of announcement-day returns with category extremeness. The design asks whether the same immediate return response is more likely to reverse when it comes from a fatter-tailed category.
- **Why this is disciplined.** The theory makes several linked predictions, not just one: expectations overreact more, prices reverse more, and trading volume is higher for more extreme categories. The empirical tests line up with all three margins.

4.2 Main threats to credibility

- **Mechanical size effects.** Perhaps categories with larger day-0 returns reverse more, independent of tails.
- **Timing and leakage.** Announcement dates may be noisy or strategically chosen.
- **Overlapping news.** Reactions to prior news may contaminate measured post-announcement returns.
- **Sample selection.** Results might depend on excluding small-cap stocks or particular categories.
- **Alternative category characteristics.** Extremeness might proxy for informativeness, media coverage, familiarity, or whether news is scheduled.

4.3 How the paper answers those threats

- **Flexible return controls.** The paper adds decile controls and cubic polynomials in day-0 returns, so the extremeness effect is not just the sign or size of the initial announcement move.
- **Alternative tail measures.** Extremeness computed from earnings growth or longer-run returns gives similar results.
- **Timing robustness.** Using a wider $[-2, +2]$ announcement window, adding hour-by-day-of-week controls, and excluding Friday announcements all leave the main coefficient negative.
- **Overlap robustness.** Excluding earnings-overlapping categories, excluding firms with any news in the previous thirty days, and directly controlling for prior announcements all preserve the result.
- **Sample robustness.** Results remain in broader category samples and in small-cap stocks.
- **Inference robustness.** The paper reports two-way clustering, Driscoll–Kraay standard errors, and a block bootstrap. The sign and magnitude of the main coefficient remain similar.
- **Horse races with alternatives.** Informativeness, media, category frequency, scheduled timing, and other category characteristics do not subsume extremeness.

4.4 What to keep in mind

- **Announcement-day returns are a proxy for fundamentals.** The empirical implementation assumes that short-run announcement returns are informative about the tail structure of the category's underlying fundamentals.
- **Categories are a coarse similarity metric.** The paper interprets categories as one important cue for associative recall, but acknowledges that investors also use text, firm history, industry, and the perceived significance of the event.
- **Underreaction is not fully eliminated.** The cross-category prediction is very strong, but the paper notes that the overall level of underreaction in the data may still reflect inattention, complexity, or capital frictions beyond the model.

5 Tables and figures

5.1 Figure 1: Diagnostic expectations and under/overreaction

Read:

- The figure is the core intuition picture. In panel (a), the category distribution is more extreme than the reference. The diagnostic posterior becomes even flatter-tailed than the rational posterior, so the mean overshoots.
- In panel (b), the category is less extreme than the reference. Then the diagnostic posterior puts relatively *less* mass on tail outcomes than the rational posterior, so expectations undershoot.
- The important insight is that the same behavioral model can generate both overreaction and underreaction depending on the shape of the category's tail.

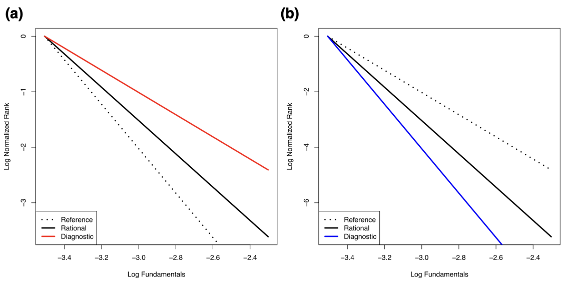

FIGURE 1

DE and under-overreaction. (a) DE, more extreme category. (b) DE, less extreme category

Notes: (a) and (b) show the DE distortions of subjective fundamentals for more and less extreme news categories, respectively. The solid red and blue curves plot the log fundamentals versus normalized log rank of the distributions of subjective fundamentals under DE. The solid and dotted black curves plot the same under the rational distributions and the reference distributions.

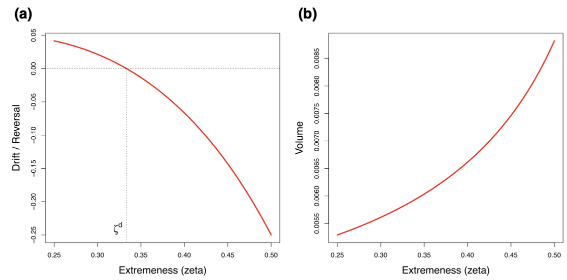

FIGURE 2

DE predictions: over-and-underreaction, volume, and extremeness. (a) Drift-reversals versus extremeness. (b) Volume versus extremeness

Notes: (a) Plots the theoretical relationship between return drift/reversal and the extremeness of the distribution of fundamentals. The dashed vertical line (ζ_d) corresponds to the extremeness of the distribution of fundamentals for the reference distribution. The dashed horizontal line corresponds to a drift/reversal coefficient β_C^{ret} of zero. (b) Plots the theoretical relationship between trading volume and the extremeness of the distribution of fundamentals. Volume is defined as half of absolute asset holdings at $t = 1$, holding fixed fundamentals.

5.2 Figure 2: Theory – reversals and volume versus extremeness

Read:

- Panel (a) shows that the theoretical drift/reversal coefficient β_C^{ret} is decreasing in extremeness and crosses zero exactly at the reference tail ζ_d.
- Panel (b) shows that announcement-day trading volume rises with extremeness because diagnostic traders disagree more when the category is more associated with tail outcomes.
- This figure is important because it makes clear that the paper is not testing a vague behavioral story. It is testing a sharp monotone prediction: more extreme categories should have more reversal and more volume.

5.3 Table 1: News category summary statistics

Read:

- Announcement days are high-volatility and high-turnover days. Earnings calls, guidance events, and credit-watch announcements are especially large in absolute returns and volume.
- The no-news benchmark has turnover 0.99, which is lower than most news categories. This is consistent with the model's view that public announcements trigger active disagreement and trading.
- The table also shows that categories differ a lot in frequency, which matters later when thinking about familiarity and scheduled-news alternative explanations.

5.4 Figure 3: Estimating extremeness

Read:

- Panel (a) shows that the tails of announcement-day returns are much closer to straight lines in log-rank/log-size space than a normal benchmark. That is exactly what a power law would imply.
- Panel (b) shows economically meaningful cross-category dispersion in estimated extremeness. The range is roughly 0.30 to 0.56.

TABLE 1

News category summary statistics

Category	N	Mean	MeanAbs	StDev	P50	Turnover
Alliance	3,326	0.15	1.29	2.19	0.09	0.97
Annual meeting	6,259	0.04	1.18	1.76	0.05	1.03
Board changes	22,173	0.06	1.35	2.14	0.08	1.08
CEO change	1,132	0.03	2.12	3.75	0.02	1.88
CFO change	1,429	−0.07	1.66	3.20	0.05	1.44
Client	20,990	0.10	1.26	1.98	0.10	0.95
Credit watch	1,323	0.28	2.38	4.00	0.10	2.19
Dividend	1,093	0.14	2.37	3.59	0.19	1.85
Downsize	2,904	−0.00	1.58	2.76	0.03	1.42
Earnings	23,573	0.13	2.59	4.01	0.12	1.97
Earnings call	15,912	0.26	3.00	4.40	0.23	2.20
Expansion	10,326	0.06	1.35	2.19	0.06	1.19
Guidance confirm	19,528	0.08	2.66	4.16	0.11	2.12
Guidance lower	1,172	−1.54	3.47	5.64	−0.64	2.53
Guidance raised	2,791	0.90	2.71	4.09	0.56	1.93
Lawsuit	5,669	0.03	1.36	2.27	0.04	1.26
M&A closing	11,294	0.10	1.29	1.98	0.08	1.01
M&A rumour	3,622	0.43	1.75	3.21	0.13	1.41
M&A transaction	5,143	0.33	1.71	3.01	0.16	1.32
No events	2801,102	0.07	1.26	2.12	0.07	0.99
Op. result	1,456	0.00	2.37	3.38	0.00	1.95
Product	23,500	0.13	1.38	2.56	0.08	1.03
Seek investment	7,900	0.15	2.01	3.22	0.09	1.43
Structure change	2,596	0.10	1.30	2.10	0.14	1.16
Write-off	3,103	0.03	2.49	3.91	0.07	1.90

Notes: Table 1 reports the summary statistics of announcement-day stock returns and trading volume for each news category in our dataset. Observations are at the news announcement level from 1 January 2011 to 31 December 2018, inclusive. The sample is all firms listed on the major U.S. stock exchanges with at least $2 billion in market capitalization. N is the number of observations. Mean is the mean, Mean of Abs. is the mean of the absolute value, StDev is the standard deviation, and P50 is the median, of announcement-day returns, respectively. Mean, Mean of Abs., StDev, and P50 are all returns measured in percentage points. Turnover is the announcement-day trading volume defined as the number of shares traded times the share price divided by the total market capitalization times 100.

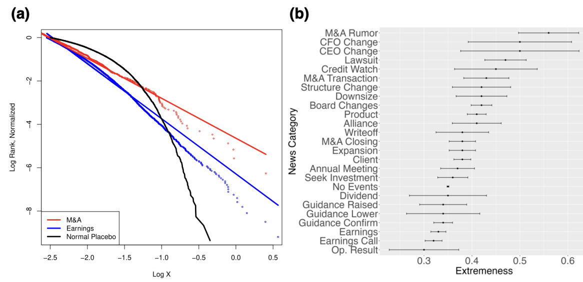

FIGURE 3

Estimating extremeness: ζ_C. (a) Extremeness estimation: example. (b) Extremeness estimates

Notes: (a) Plots the estimates corresponding to equation (10) for M&A events (top line), earnings events (bottom line), and a simulated normally distributed return distribution (curved line). The *x*-axis shows the normalized log value of absolute announcement-day returns, while the *y*-axis shows the normalized log rank. The solid lines plot the linear best fit corresponding to equation (10). (b) Plots the extremeness ζ_C estimates for each category corresponding to equation (10). 95% CIs are computed following Gabaix and Ibragimov (2011).

- Categories such as M&A rumor, CEO change, CFO change, and lawsuit are among the most extreme; operating results, earnings calls, and earnings are among the least extreme.
- This figure validates the key empirical premise of the paper: categories differ in tail thickness in a measurable way.

5.5 Table 2: Category-level extremeness and drift/reversal

Read:

- Part (a) pairs each category's estimated extremeness ζ_C with its empirical drift/reversal coefficient β_C.
- The most negative β_C's occur in very extreme categories: CEO change ($\beta = -0.42$), CFO change (-0.31), M&A rumor (-0.26), M&A transaction (-0.25), and structure change (-0.26).
- Categories with lower extremeness are closer to drift or weak continuation: operating results ($\beta = 0.21$), annual meetings (0.19), guidance confirm (0.12), and earnings-related news (0.06 to 0.07).
- Part (b) rejects both "all categories have the same β_C" and "all $\beta_C = 0$," even after adding industry fixed effects and flexible controls for announcement returns. So there is genuine category-level heterogeneity to explain.

5.6 Table 3: Main regression of reversals on extremeness

Read:

- This is the headline table. Across all six specifications, the interaction between announcement-day returns and extremeness is negative and strongly significant.
- In the baseline, the coefficient on $r_{i,t} \times \zeta_C$ is -1.70. With market-benchmarked returns, it is -1.44. With time-varying tails and benchmarking, it remains around -1.83 to -1.85.

TABLE 2
Category-level heterogeneity

Category	ζ	ζ s.e.	β	β s.e.
(a) Category-level extremeness and drift/reversal				
Alliance	0.41	0.026	−0.13	0.23
Annual meeting	0.37	0.018	0.19	0.20
Board changes	0.42	0.011	−0.07	0.08
CEO change	0.50	0.063	−0.42	0.19
CFO change	0.50	0.055	−0.31	0.31
Client	0.38	0.009	−0.18	0.09
Credit watch	0.45	0.044	0.04	0.20
Dividend	0.35	0.041	−0.05	0.13
Downsize	0.42	0.027	0.12	0.20
Earnings	0.33	0.008	0.07	0.04
Earnings call	0.32	0.009	0.06	0.04
Expansion	0.38	0.014	−0.09	0.10
Guidance confirm	0.34	0.010	0.12	0.04
Guidance lower	0.34	0.039	−0.09	0.12
Guidance raised	0.34	0.025	0.06	0.09
Lawsuit	0.47	0.022	0.12	0.15
M&A closing	0.38	0.014	0.08	0.12
M&A rumour	0.56	0.032	−0.26	0.15
M&A transaction	0.43	0.024	−0.25	0.11
Op. result	0.30	0.037	0.21	0.24
Product	0.41	0.010	0.03	0.09
Seek investment	0.36	0.016	0.12	0.07
Structure change	0.42	0.031	−0.26	0.27
Write-off	0.38	0.028	0.13	0.11
(b) Testing for heterogeneity				
Hypothesis	(1)	(2)	(3)	(4)
Jointly equal	1.79	3.35	1.77	1.87
p-value	0.0042	<0.0001	0.0053	0.0021
Jointly equal to 0	2.55	1.84	1.72	2.92
p-value	<0.0001	0.0030	0.0072	<0.0001
Industry FEs		X		X
Return controls			X	X

Notes: Table 2(a) reports the extremeness $\zeta_{\mathcal{C}}$ and $\beta_{\mathcal{C}}$, the drift/reversal estimates, for each news category. ζ is the extremeness, *i.e.* inverse power-law index estimated using equation (10). ζ s.e. is the standard error for $\zeta_{\mathcal{C}}$ and is computed following Gabaix and Ibragimov (2011). $\beta_{\mathcal{C}}$ is the post-announcement drift/reversal beta estimated using equation (13). $\beta_{\mathcal{C}}$ s.e. is the standard error for $\beta_{\mathcal{C}}$, two-way clustered at the firm and day levels. Table 2(b) reports the *F*-statistics for equation (14). Jointly equal is the hypothesis that all $\beta_{\mathcal{C}}$'s are equal. Jointly equal to 0 is the hypothesis that all $\beta_{\mathcal{C}}$'s are equal to 0. Industry FEs are industry-fixed effects and interaction terms of dummy variables for each industry with the announcement-day return. Return Controls are cubic polynomials of the announcement-day returns.

TABLE 3
Reversals and extremeness

	(1)	(2)	(3)	(4)	(5)	(6)
Announcement-day return	0.62^{***}	0.56^{***}	0.56^{***}	0.82^{***}	0.68^{***}	0.68^{***}
	(0.19)	(0.17)	(0.17)	(0.27)	(0.25)	(0.25)
Announcement-day return × Extremeness	-1.70^{***}	-1.44^{***}	-1.44^{***}	-2.22^{***}	-1.83^{***}	-1.85^{***}
	(0.53)	(0.47)	(0.47)	(0.74)	(0.69)	(0.69)
Extremeness			−0.001			0.02
			(0.02)			(0.03)
Constant	0.02^{***}	-0.01^{***}	−0.01	0.01^{***}	-0.01^{***}	-0.02^{**}
	(0.003)	(0.002)	(0.01)	(0.003)	(0.002)	(0.01)
Time-varying tails	No	No	No	Yes	Yes	Yes
Return benchmark	No	Yes	Yes	No	Yes	Yes
Observations	197,498	197,498	197,498	110,748	110,748	110,748

Notes: Table 3 reports the estimates corresponding to equation (15). Observations are at the news announcement level from 1 January 2011 to 31 December 2018. The dependent variable is the cumulative post-announcement return from day 1 to day 90 after the announcement. Announcement-day return is the stock return of the firm on the day of the announcement and is measured in percentage points. Extremeness is the inverse power-law index $\zeta_{\mathcal{C}}$ estimated following equation (10). Time-varying tails indicates whether Extremeness is computed over a rolling past 5-year window (Yes) or over the entire sample (No). Return benchmark indicates whether the Announcement-day return and dependent variable are excess returns relative to the S&P 500 (Yes) or raw returns (No). Standard errors are two-way clustered at the firm and day levels. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

- The stability across columns matters. The result is not coming from future information in tail estimation, from raw-market movements, or from omitting the main effect of extremeness.
- Quantitatively, the implied post-announcement slope moves from modest drift in low-ζ categories to sizable reversal in high-ζ categories.

5.7 Figure 4: Category-level scatter and horizon dynamics

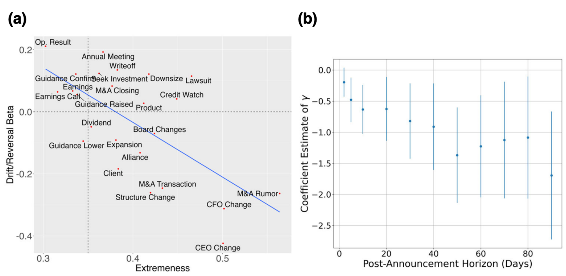

FIGURE 4
Extreme categories and reversals. (a) $\beta_{\mathcal{C}}$ versus $\zeta_{\mathcal{C}}$. (b) γ over time

Notes: (a) Plots the relationship between extremeness and post-announcement drift/reversal $\beta^{\text{ret}}_{\mathcal{C}}$ for each news category $\mathcal{C}$. Extremeness is the inverse power-law index $\zeta_{\mathcal{C}}$ estimated following equation (10). Drift/Reversal Beta is the post-announcement drift or reversal coefficients $\beta^{\text{ret}}_{\mathcal{C}}$ estimated following equation (13). The dotted horizontal line indicates where drift/reversal $\beta^{ret}_{\mathcal{C}} = 0$. The dotted vertical line indicates where $\zeta_{\mathcal{C}} = 0.35$, which is the extremeness of the No News distribution. (b) Plots the estimated γ over different post-announcement horizons from $k = 2$ to $k = 90$, with the γ for each horizon k being estimated following equation (15). The vertical lines plot the 95% CIs for each coefficient estimate.

TABLE 4
Summary of robustness exercises

	Specification	Coefficient	SE	Observations
1	Baseline	−1.7	0.52	197,498
2	Earnings growth extremeness	−1.21	0.46	197,498
3	Long-run return extremeness	−1.38	0.64	197,498
4	Two-day announcement window	−2.82	0.33	197,498
5	Hour by day of week controls	−1.7	0.52	197,498
6	Exclude Friday announcements	−1.94	0.57	174,900
7	Non-parametric return controls (deciles)	−1.75	0.52	197,498
8	Non-parametric return controls (polynomial)	−1.51	0.55	197,498
9	Exclude smallest 25% of news	−1.67	0.53	148,126
10	Exclude smallest 50% of news	−1.74	0.53	98,750
11	Exclude top 0.01% of news	−1.45	0.55	197,300
12	Exclude top 1% of news	−1.6	0.66	195,522
13	Positive news only	−2.19	0.85	104,783
14	Negative news only	−0.62	1.25	92,715
15	Exclude earnings-overlapping categories	−1.67	0.54	152,513
16	Exclude news within 30 days	−3.23	0.92	35,879
17	Overlapping news controls	−1.43	0.45	197,498
18	All news categories	−1.19	0.48	250,852
19	All news categories with 1,000+ occurrences	−1.27	0.5	243,966
20	Small-cap stocks	−1.22	0.41	226,986
21	Post-announcement horizon $k = 30$-day	−0.82	0.31	197,498
22	Post-announcement horizon $k = 60$-day	−1.23	0.42	197,498
23	Exclude attrited firms	−1.7	0.52	196,463
24	Exclude high-attrition news categories	−1.91	0.54	142,651
25	Driscoll–Kraay standard errors	−1.7	0.54	197,498
26	Block bootstrap	−0.92	0.33	197,498

Notes: Table 4 summarizes the robustness exercises for equation (15). Coefficient is the main γ coefficient estimate. SE is the standard error. Observations are the number of observations in the corresponding estimates for each row. Row 1 is the baseline estimate. Rows 2 and 3 use alternative measures of extremeness. Row 4 uses ±2 days as the announcement window. Row 5 includes hour-by-day-of-week controls. Row 6 excludes Friday announcements. Rows 7 and 8 include decile and cubic polynomial functions of announcement-day returns as controls. Rows 9–12 exclude the smallest 25% and 50%, and largest 0.01% and 1% of news by absolute announcement-day returns, respectively. Rows 13 and 14 report estimates on news with positive and negative announcement-day returns only, respectively. Rows 15 and 16 exclude news categories that overlap with other news. Row 17 includes controls for overlapping news as in equation (16). Rows 18 and 19 reports the estimates for all news categories and categories with 1,000+ occurrences. Row 20 reports the results on small-cap firms. Rows 21 and 22 set the post-announcement horizons k as 30 and 60 days. Row 23 excludes announcements by firms that attrited during our sample. Row 24 excludes news categories that had above-average attrition rates. Row 25 computes Driscoll and Kraay (1998) standard errors. Row 26 uses a block bootstrap approach (Politis and Romano, 1994) on a full firm-day panel to account for compounded estimation errors.

Read:

- Panel (a) is the empirical mirror image of Figure 2a. There is a clear negative relationship between ζ_C and β_C, with correlation around -0.65.
- Panel (b) shows that the estimated γ is negative across horizons from 2 to 90 days, with most of the reversal materializing by about day 40.
- This is useful because it shows the pattern is not a one-day bounce. It is a short-horizon correction that builds over several weeks.

5.8 Table 4: Robustness exercises

Read:

- Nearly every row delivers the same qualitative conclusion: the main interaction coefficient stays negative.
- The result survives alternative extremeness measures, wider announcement windows, exclusion of Friday announcements, flexible controls for return magnitude, deletion of small news and outliers, exclusion of overlapping-news categories, alternative horizons, and alternative standard-error methods.
- Two especially reassuring rows are the overlap controls and the block bootstrap. They show the finding is not just a by-product of contaminated return windows or generated-regressor issues.

5.9 Table 5 and Figure 5: Volume and extremeness

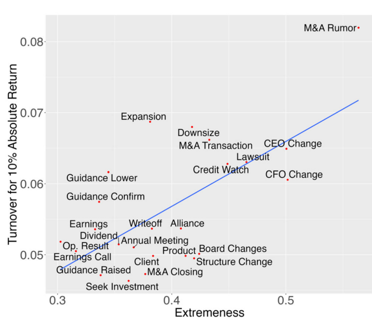

FIGURE 5

Volume and extremeness

Notes: Plots the relationship between extremeness and conditional average turnover for a news announcement with a 10% absolute announcement-day return $\overline{Turnover}_{\mathcal{C},10}$ for each news category $\mathcal{C}$. Extremeness is the inverse power-law index $\zeta_{\mathcal{C}}$ estimated following equation (10). Turnover for 10% Absolute Return is the conditional average turnover for a news announcement with a 10% absolute announcement-day return, $\overline{Turnover}_{\mathcal{C},10}$, estimated following equation (18).

TABLE 5

Volume and extremeness

Variables	(1) Turnover	(2) Turnover	(3) Turnover	(4) Turnover
Abs. Announcement-day return	0.25^{***}	0.22^{***}	0.19^{**}	0.18^{**}
	(0.084)	(0.081)	(0.087)	(0.085)
Abs. Announcement-day return × Extremeness	0.65^{***}	0.78^{***}	0.93^{***}	0.94^{***}
	(0.23)	(0.23)	(0.24)	(0.24)
Constant	0.0053^{***}	0.0050^{***}	0.0058^{***}	0.0058^{***}
	(0.00028)	(0.00031)	(0.00026)	(0.00028)
Observations	197,498	197,497	197,498	197,497
R-squared	0.371	0.394	0.395	0.412
News category FEs	Yes	Yes	Yes	Yes
Trading day FEs	No	Yes	No	Yes
Return benchmark	No	No	Yes	Yes

Notes: Table 5 reports the estimates corresponding to equation (17). Observations are at the news announcement level from 1 January 2011 to 31 December 2018. The dependent variable is the announcement-day turnover, defined as the volume of shares traded times the share price divided by the market capitalization. Abs. Announcement-day return is the absolute value of the announcement-day return and is measured in percentage points. Extremeness is the inverse power-law index $\zeta_{\mathcal{C}}$ estimated following equation (10). Trading day FEs indicates whether the specification includes trading day fixed effects. Return benchmark indicates whether the Announcement-day return and dependent variable are excess returns relative to the S&P 500 (Yes) or raw returns (No). Standard errors are two-way clustered at the firm and day levels. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Read:

- Table 5 tests Prediction 3. The interaction between absolute announcement-day returns and extremeness is positive in every specification, ranging from 0.65 to 0.94.
- Figure 5 shows the same result visually: categories with larger ζ_C have higher fitted turnover for a 10% absolute return shock.
- This is important because it supports the paper's mechanism, not just the price-pattern result. Extreme categories induce more disagreement and more trading, which is what the diagnostic-expectations story implies.

5.10 Table 6 and Figure 6: Alternative explanations

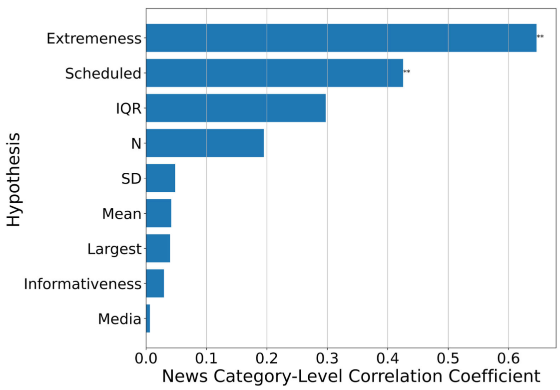

FIGURE 6
Comparison of explanatory power

Notes: Plots the category-level correlation coefficients between the category's post-announcement drift or reversal coefficient $\beta_{\mathcal{C}}$ and either extremeness or other alternative explanatory variables (Hypothesis). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

TABLE 6
Alternative explanations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Announcement-day return	0.615^{***}	0.917^{***}	0.816^{***}	0.609^{***}	0.700^{***}	0.691^{***}	0.984^{***}	0.611^{***}	0.614^{**}
	(0.185)	(0.294)	(0.232)	(0.189)	(0.204)	(0.214)	(0.300)	(0.185)	(0.246)
Announcement-day return × Extremeness	-1.593^{***}	-2.081^{***}	-1.836^{***}	-1.479^{***}	-1.814^{***}	-1.633^{***}	-2.419^{***}	-1.559^{***}	-1.586^{***}
	(0.498)	(0.603)	(0.505)	(0.525)	(0.548)	(0.510)	(0.734)	(0.573)	(0.548)
Announcement-day return × Alternative var.	2.121	−0.039	−2.803	−0.245	-0.024^{*}	−0.147	-0.109^{*}	−0.000	−0.061
	(7.207)	(0.032)	(2.432)	(0.168)	(0.014)	(0.161)	(0.065)	(0.000)	(8.061)
Alternative hypothesis	Mean	IQR	SD	Abs.	*N*	Largest	Scheduled	Media	Informativeness
Observations	197,498	197,498	197,498	197,498	197,498	197,498	197,498	197,498	197,498

Notes: Table 6 reports the estimates corresponding to equation (19). Observations are at the news announcement level from 1 January 2011 to 31 December 2018. The dependent variable is the post-announcement return, *i.e.* cumulative return from day 1 to day 90 subsequent to the announcement. Announcement-day return is the return of the stock on the day of the announcement, expressed in percentage points. Extremeness is the inverse power-law index $\zeta_{\mathcal{C}}$ estimated following equation (10). Alternative var. is the explanatory regressor for alternative hypotheses, computed at the category level. Mean is the average announcement-day return. IQR is the interquartile range of the announcement-day return. SD is the standard deviation of the announcement-day return. Abs. is the absolute value of the announcement-day return. N is the number of total occurrences of the announcement. Largest is the average of the five largest absolute announcement-day returns for each news category. Scheduled is an indicator variable for news categories that occur on pre-announced schedules (operating results, earnings, guidances, dividends, earnings calls, and annual meetings). Media is the average number of news articles written about the firm on the day of the announcement. Informativeness is the measure of price informativeness computed following Dávila and Parlatore (2018). Standard errors are two-way clustered at the firm and day levels. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Read:

- Table 6 runs horse races between extremeness and alternative category-level explanations such as mean return, IQR, standard deviation, absolute-return magnitude, frequency, largest announcements, scheduled timing, media, and informativeness.
- Extremeness remains negative and significant in all columns, which means it is not simply proxying for any one of those other characteristics.
- Figure 6 shows the category-level predictive ranking: extremeness has by far the strongest correlation with the drift/reversal coefficients. Scheduled news comes second, but still far behind.
- The key message is that what matters is not just how informative or how public the news is, but whether the category is associated with salient tail events.

6 Short summary

- **Conclusion.** News from more extreme categories is more likely to be overreacted to, and this overreaction shows up as stronger post-announcement reversals and higher trading volume.
- **Intuition.** Extreme categories cue memorable outlier events. Diagnostic investors then overweight tail states and push prices too far on the announcement day.
- **Empirical lesson.** When trying to explain drift versus reversal, it is not enough to look only at the sign or size of the news. The tail structure of the broader category contains powerful information about how investors will process it.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Category-level extremeness predicts category-level overreaction.

- **Magnitude.** In the data, the least-extreme categories exhibit modest drift, while the most-extreme categories exhibit reversals of around one-quarter of the announcement-day return.
- **Mechanism evidence.** More extreme categories also generate higher announcement-day turnover, consistent with greater disagreement when investors overweight salient tail outcomes.
- **Expectations evidence.** Analyst-forecast evidence points in the same direction, though the paper is cautious and treats it as suggestive.
- **Robustness.** The result survives changes in how tails are estimated, how returns are benchmarked, how event windows are defined, how overlapping news is handled, and how inference is conducted.

7.2 Economic intuition

- Investors do not process news in a vacuum. They map it into a category and recall comparable episodes.
- Categories with fat tails make their rare outliers unusually salient. That changes the subjective distribution investors attach to new announcements from that category.
- Because rational arbitrageurs arrive only gradually in the model, prices initially reflect those distorted category-conditioned beliefs and later mean-revert.

7.3 Takeaway

The paper's contribution is to show that a simple statistical object – the tail thickness of a news category – can organize a large amount of heterogeneity in market reactions to news. It unifies post-earnings drift and short-run reversals within a single framework: news from thin-tailed categories tends to underreact, while news from fat-tailed categories tends to overreact. More broadly, the paper is a strong example of how a behavioral idea becomes much more persuasive once it is turned into a measurable cross-sectional predictor with multiple linked empirical implications.